

Optimization

3.2.3 Nelder–Mead simplex method

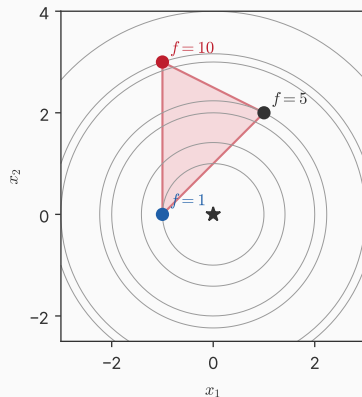
José Firmino Aguilar Madeira

From the grid to the simplex: can we use what we have already evaluated?

The grid (3.2.2) finds promising regions, but keeps spending evaluations where we already know it does not matter: as in random search, the points **do not use what has already happened**.

In 2D, with $f(x_1, x_2) = x_1^2 + x_2^2$, we evaluated three points:

$$f(-1, 0) = 1, \quad f(1, 2) = 5, \quad f(-1, 3) = 10.$$



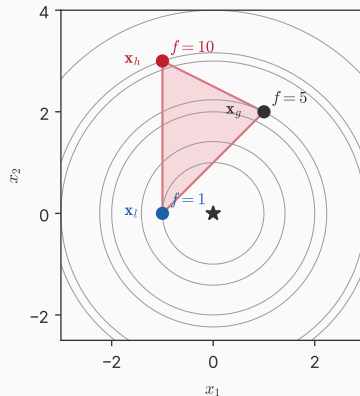
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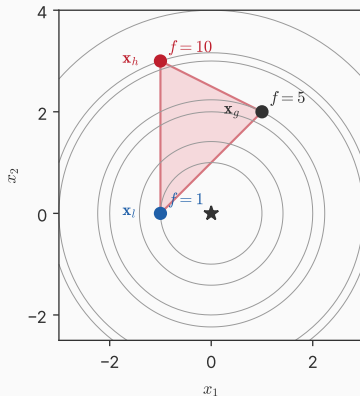
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If we only knew these three values, where would we try next?



Three points in 2D: reflection is born

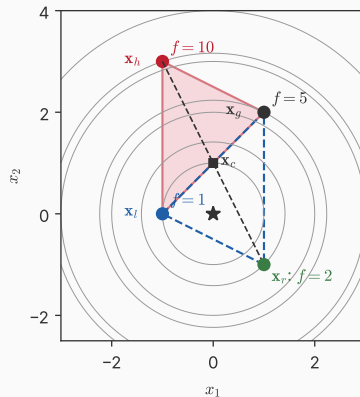
The natural answer: **on the side opposite the worst point.**

Centre of the other two: $\mathbf{x}_c = \frac{\mathbf{x}_l + \mathbf{x}_g}{2} = (0, 1)$.

Reflect the worst through $\mathbf{x}_c$:

$$\boxed{\mathbf{x}_r = \mathbf{x}_c + (\mathbf{x}_c - \mathbf{x}_h)} = 2\mathbf{x}_c - \mathbf{x}_h = (1, -1), \quad f(\mathbf{x}_r) = 2.$$

The worst ($f = 10$) is replaced by a point with $f = 2$:
the triangle **flips** away from the worst vertex.



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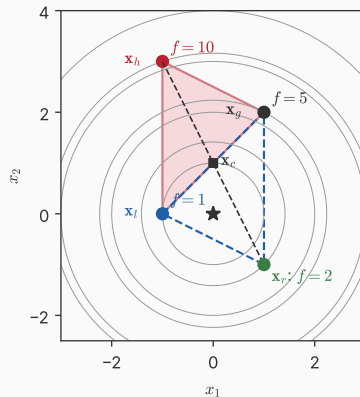
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The central idea of Nelder–Mead

With a few values of f , the **geometry** of the points already evaluated tells us where to try next. No gradient was computed.



The simplex in n dimensions

A **simplex** in $\mathbb{R}^n$ is the polytope with $n + 1$ vertices:

$n = 1$: segment; $n = 2$: triangle; $n = 3$: tetrahedron.

Nondegenerate = with nonzero (n -dimensional) volume (the vertices do not all lie in a hyperplane).

Nelder–Mead keeps $n + 1$ points and uses **only their values of f** . At each iteration it:

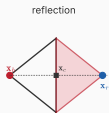
1. identifies the best $\mathbf{x}_l$, the next to the worst $\mathbf{x}_g$ and the worst $\mathbf{x}_h$;
2. computes the centroid of the n vertices that are *not* the worst: $\mathbf{x}_c = \frac{1}{n} \sum_{i \neq h} \mathbf{x}_i$;
3. tries points on the line $\mathbf{x}_h \rightarrow \mathbf{x}_c$, on the other side.

Why it works

With $n + 1$ values of f the simplex “sees” the local slope without gradients: it moves away from the worst and heads in the direction in which f seems to decrease.

In 2D, $n = 2$: the centroid of the two best is the midpoint, as on the previous page.

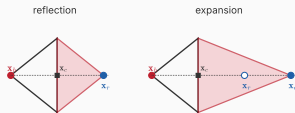
And if the reflection is very good, or bad?



All these points lie on the line defined by x_h and x_c .

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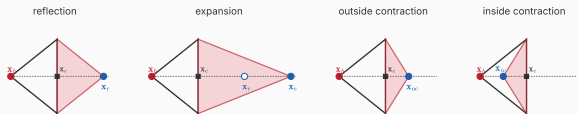
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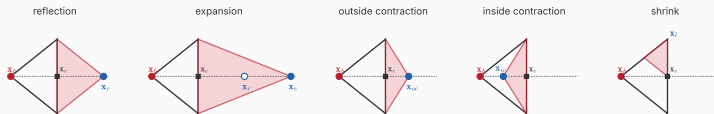
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 $x_{oc} = (1 + \beta)x_c - \beta x_h$; or inside, between x_c and x_h :
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- **Not even contraction works?** The simplex is too large or badly placed: **shrink** towards x_l , setting $x_i = x_l + \frac{1}{2}(x_i - x_l)$.

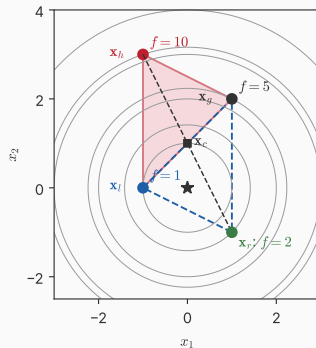
Operation	When
Expansion	$f(x_r) < f(x_l)$
Reflection accepted	$f(x_l) \leq f(x_r) < f(x_g)$
Outside contraction	$f(x_g) \leq f(x_r) < f(x_h)$
Inside contraction	$f(x_r) \geq f(x_h)$

$\gamma > 1$ (expansion), $0 < \beta < 1$ (contraction); usual values $\gamma = 2$, $\beta = \frac{1}{2}$ (Nelder and Mead, 1965). The same formulas can be written $x_e = x_c + \gamma(x_r - x_c)$, $x_{oc} = x_c + \beta(x_r - x_c)$, $x_{ic} = x_c - \beta(x_c - x_h)$.

One iteration by hand

$f(x_1, x_2) = x_1^2 + x_2^2$; the triangle from the start.

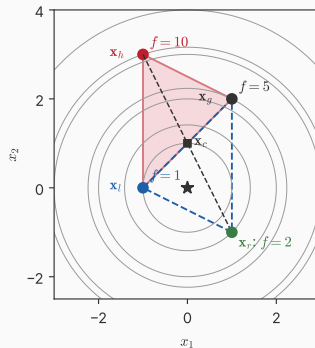
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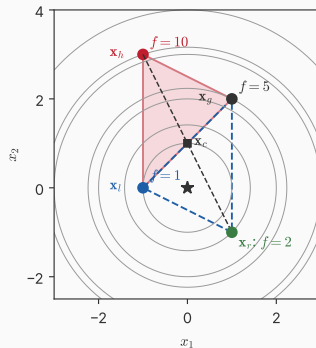
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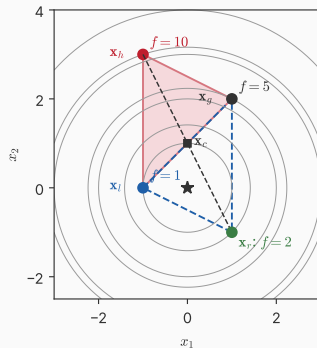
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2. **Step 3:** $\mathbf{x}_r = 2\mathbf{x}_c - \mathbf{x}_h = (0, 2) - (-1, 3) = (1, -1)$; $f(\mathbf{x}_r) = 2$. $f(\mathbf{x}_l) = 1 \leq f(\mathbf{x}_r) = 2 < f(\mathbf{x}_g) = 5 \Rightarrow$ reflection **accepted** (it is not better than the best, so no expansion; it is better than the next to the worst, so no contraction).



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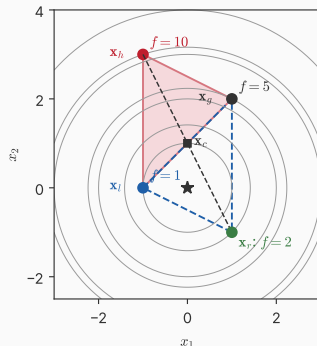
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3. **New simplex:** $\{(-1, 0), (1, -1), (1, 2)\}$, with $f = 1, 2, 5$. At the next iteration, the worst is $(1, 2)$.



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At the next iteration, the worst is $(1, 2)$.
4. **Step 4:**
$$Q = \left(\frac{(1-1)^2 + (2-1)^2 + (5-1)^2}{3} \right)^{1/2} = \sqrt{17/3} = 2.380$$
: still
large, continue. It cost **two** evaluations of f ($\mathbf{x}_c$ and $\mathbf{x}_r$).



The new triangle already contains the minimum $(0, 0)$; the following iterations will shrink it around the minimum.

Nelder–Mead simplex ($\gamma > 1$, $0 < \beta < 1$, $\varepsilon > 0$)

Step 1 Choose γ , β , ε and an initial simplex with $n + 1$ vertices; evaluate f at the vertices.

Step 2 Identify $\mathbf{x}_l$ (best), $\mathbf{x}_g$ (next to the worst) and $\mathbf{x}_h$ (worst); compute $\mathbf{x}_c = \frac{1}{n} \sum_{i \neq h} \mathbf{x}_i$ and $f(\mathbf{x}_c)$.

Step 3 Compute $\mathbf{x}_r = 2\mathbf{x}_c - \mathbf{x}_h$ and $f(\mathbf{x}_r)$; set $\mathbf{x}_{new} = \mathbf{x}_r$.

- If $f(\mathbf{x}_r) < f(\mathbf{x}_l)$: set $\mathbf{x}_{new} = (1 + \gamma)\mathbf{x}_c - \gamma\mathbf{x}_h$ (expansion; if $f(\mathbf{x}_{new}) \geq f(\mathbf{x}_r)$, keep $\mathbf{x}_r$)
- If $f(\mathbf{x}_r) \geq f(\mathbf{x}_h)$: set $\mathbf{x}_{new} = (1 - \beta)\mathbf{x}_c + \beta\mathbf{x}_h$ (inside contraction)
- If $f(\mathbf{x}_g) \leq f(\mathbf{x}_r) < f(\mathbf{x}_h)$: set $\mathbf{x}_{new} = (1 + \beta)\mathbf{x}_c - \beta\mathbf{x}_h$ (outside contraction)

Compute $f(\mathbf{x}_{new})$ and replace $\mathbf{x}_h$ by $\mathbf{x}_{new}$. If a contraction does not improve, shrink: set $\mathbf{x}_i = \mathbf{x}_l + \frac{1}{2}(\mathbf{x}_i - \mathbf{x}_l)$, $i \neq l$.

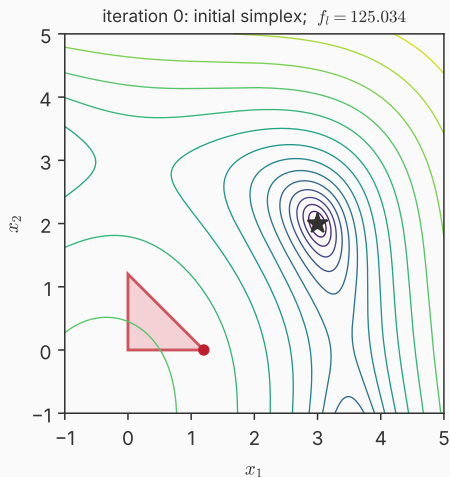
Step 4 If $Q = \left(\sum_{i=1}^{n+1} \frac{(f(\mathbf{x}_i) - f(\mathbf{x}_c))^2}{n+1} \right)^{1/2} \leq \varepsilon$, **stop**; otherwise, go to Step 2.

Error analysis and cost

- **Error analysis:** Q measures the spread of the values of f at the vertices around $f(\mathbf{x}_c)$. Near a minimum the vertices have almost equal values of f and $Q \rightarrow 0$.
- A small Q does not guarantee a small simplex: in a flat valley the values of f can be similar while the simplex is still large.
- **Cost:** $n_f = 2$ to 3 per iteration ($\mathbf{x}_c$, $\mathbf{x}_r$ and possibly $\mathbf{x}_{new}$); n more with a shrink.
- **Initial simplex:** given in the exercise; in the course code, by default, $\mathbf{x}_0$ and $\mathbf{x}_0 + h_i \mathbf{e}_i$, with $h_i = 0.05 \max(1, |x_{0,i}|)$.

sort $\rightarrow$ centroid $\rightarrow$ reflect
excellent $\rightarrow$ expand; reasonable $\rightarrow$
accept;
bad $\rightarrow$ contract; still bad $\rightarrow$ shrink
 $Q \leq \varepsilon \rightarrow$ stop

Himmelblau from (0,0): the simplex in action

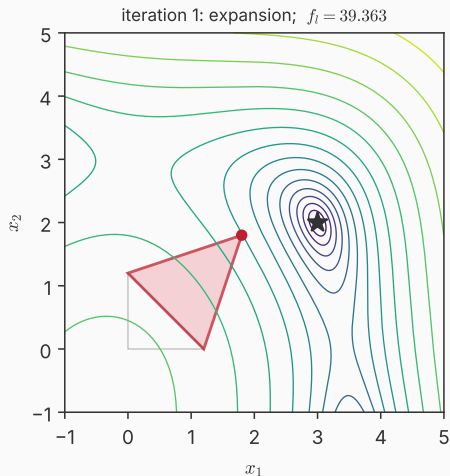


Initial simplex (0, 0), (1.2, 0), (0, 1.2).

k	operation	$\mathbf{x}_l$	$f(\mathbf{x}_l)$	$f(\mathbf{x}_h)$	Q
0	initial	(1.200, 0.000)	125.0336	170.000	–

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

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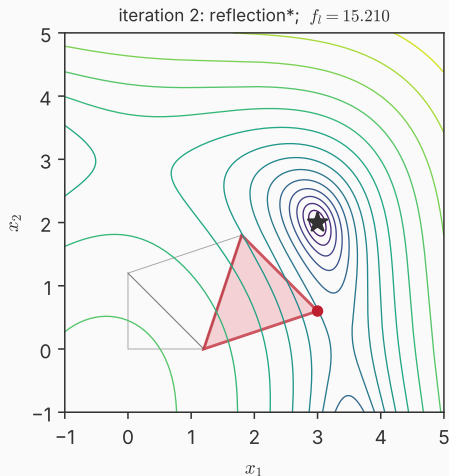


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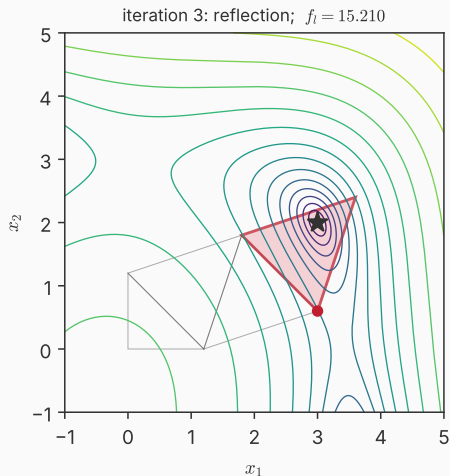


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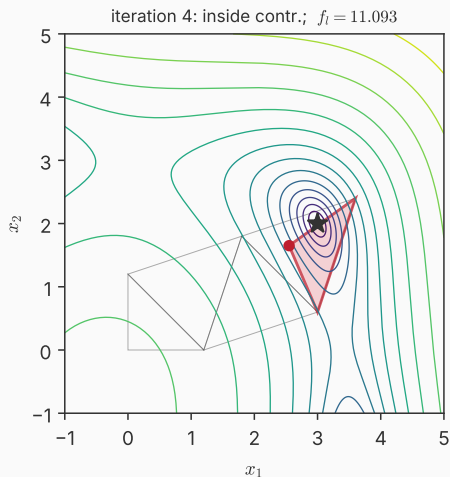


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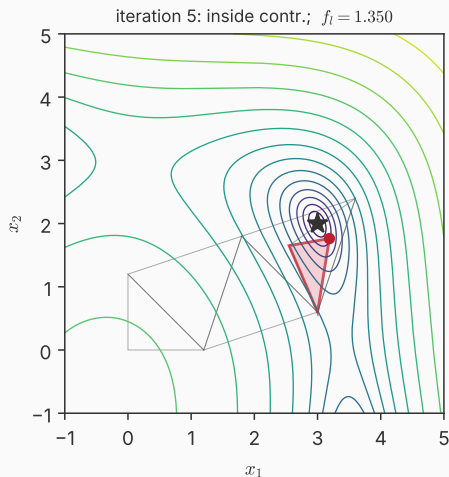


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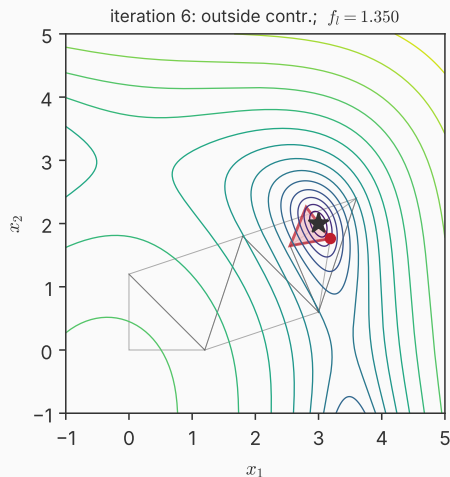


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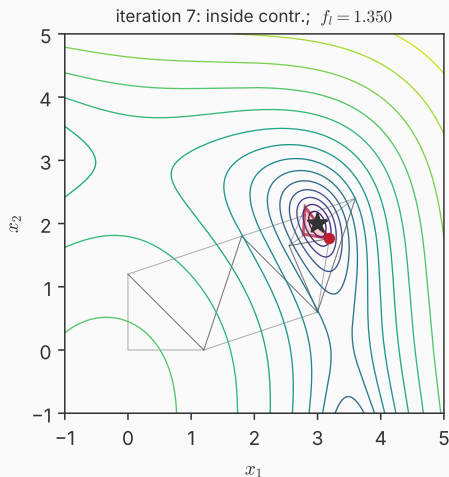


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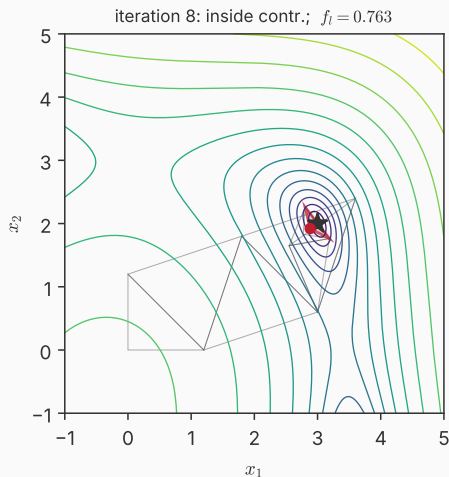


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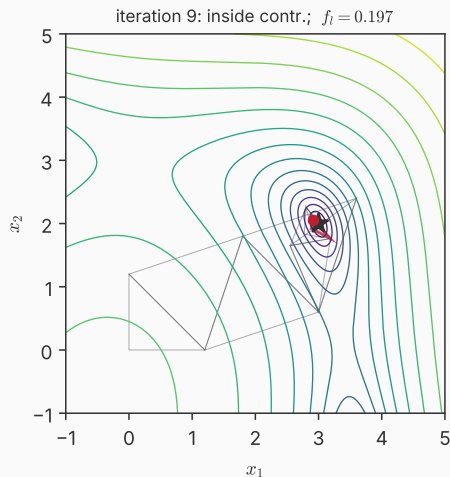


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1	expansion	(1.800, 1.800)	39.3632	126.954	57.286
2	reflection*	(3.000, 0.600)	15.2096	125.034	52.768
3	reflection	(3.000, 0.600)	15.2096	39.363	9.943
4	inside contr.	(2.550, 1.650)	11.0925	24.579	14.104
5	inside contr.	(3.188, 1.762)	1.3499	15.210	7.211
6	outside contr.	(3.188, 1.762)	1.3499	11.093	4.981
7	inside contr.	(3.188, 1.762)	1.3499	2.965	2.095
8	inside contr.	(2.884, 1.921)	0.7628	1.604	1.286

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action

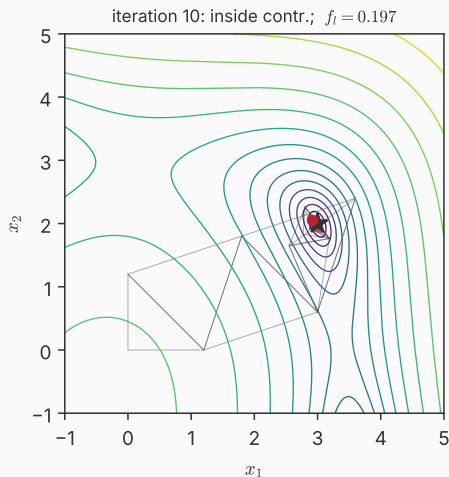


Initial simplex (0,0), (1.2,0), (0,1.2).

k	operation	$\mathbf{x}_l$	$f(\mathbf{x}_l)$	$f(\mathbf{x}_h)$	Q
0	initial	(1.200, 0.000)	125.0336	170.000	–
1	expansion	(1.800, 1.800)	39.3632	126.954	57.286
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8	inside contr.	(2.884, 1.921)	0.7628	1.604	1.286
9	inside contr.	(2.919, 2.050)	0.1972	1.350	0.643

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action

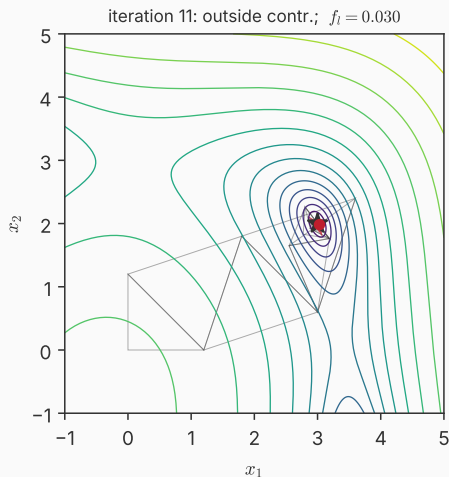


Initial simplex (0,0), (1.2,0), (0,1.2).

k	operation	$\mathbf{x}_l$	$f(\mathbf{x}_l)$	$f(\mathbf{x}_h)$	Q
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9	inside contr.	(2.919, 2.050)	0.1972	1.350	0.643
10	inside contr.	(2.919, 2.050)	0.1972	0.763	0.262

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action

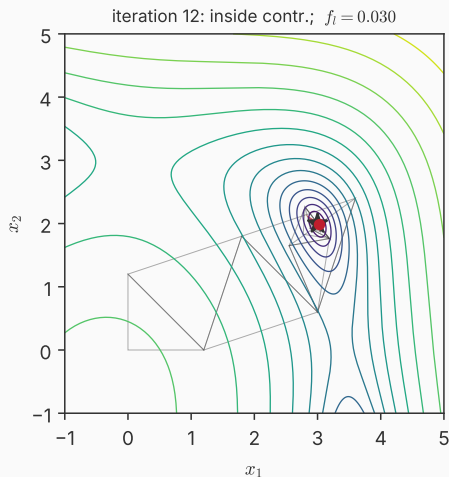


Initial simplex $(0, 0)$, $(1.2, 0)$, $(0, 1.2)$.

k	operation	$\mathbf{x}_l$	$f(\mathbf{x}_l)$	$f(\mathbf{x}_h)$	Q
0	initial	(1.200, 0.000)	125.0336	170.000	–
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9	inside contr.	(2.919, 2.050)	0.1972	1.350	0.643
10	inside contr.	(2.919, 2.050)	0.1972	0.763	0.262
11	outside contr.	(3.031, 1.983)	0.0303	0.217	0.130

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action



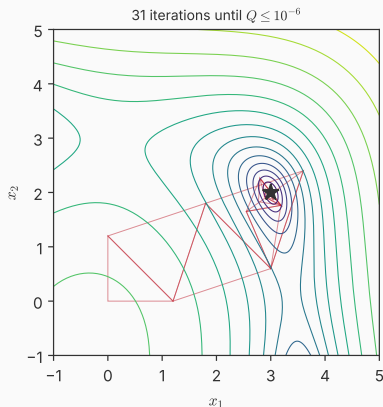
Initial simplex $(0, 0)$, $(1.2, 0)$, $(0, 1.2)$.

k	operation	$\mathbf{x}_l$	$f(\mathbf{x}_l)$	$f(\mathbf{x}_h)$	Q
0	initial	(1.200, 0.000)	125.0336	170.000	–
1	expansion	(1.800, 1.800)	39.3632	126.954	57.286
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8	inside contr.	(2.884, 1.921)	0.7628	1.604	1.286
9	inside contr.	(2.919, 2.050)	0.1972	1.350	0.643
10	inside contr.	(2.919, 2.050)	0.1972	0.763	0.262
11	outside contr.	(3.031, 1.983)	0.0303	0.217	0.130
12	inside contr.	(3.031, 1.983)	0.0303	0.197	0.104

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

The triangle **grows** when it finds descent (expansion), **flips** following the terrain (reflections) and **shrinks** near the minimum (contractions). $f(\mathbf{x}_h)$ and Q decrease in jumps.

The complete path; on the computer



Converges to (3, 2), $f \approx 0$: 31 iterations, $n_f = 92$
(with $\varepsilon = 10^{-6}$).

- Himmelblau has 4 minima: starting from (0, 0) with the simplex to the right and upwards we reach (3, 2); with the simplex (0, 0), (-1.2, 0), (0, -1.2) we reach (-2.81, 3.13). **Nelder-Mead is a local method**; the result depends on the initial simplex.

In MATLAB

`[x, fval] = fminsearch(f, x0)` is Nelder-Mead.
Useful options: `optimset('TolX', 1e-6, 'TolFun', 1e-6, 'Display', 'iter')`.

THE SAME IN PYTHON

```
r = scipy.optimize.minimize(f, x0,  
method='Nelder-Mead'); r.nfev counts the evaluations.
```

`fminsearch` and SciPy use another stopping test (tolerances on x and on f , without computing $f(\mathbf{x}_c)$) and another initial simplex: same idea, different counts.

The price of Nelder–Mead

Advantages

- uses only values of f (order 0);
- simple; 2–3 evaluations per iteration (n more with a shrink);
- adapts the position, size and shape of the simplex to the terrain;
- widely used in practice: `fminsearch`.

Rosenbrock from $(-1.5, 2)$, simplex with edges 0.5:
 $f < 10^{-4}$ with $n_f = 244$ (166 without the evaluations of $f(\mathbf{x}_c)$, as in `fminsearch`); localized random search needed 975 [867, 1052] (median [Q1, Q3], 30 runs).

Dimension: $n \lesssim 10$ is a common rule of thumb, not a boundary — it depends greatly on the problem.

Limitations

- local;
- depends on the initial simplex (on the same Rosenbrock, with 5% edges: $n_f = 211$);
- mostly used in low/moderate dimension: performance tends to degrade as n grows;
- it can stagnate: the simplex degenerates or shrinks away from the minimum;
- Nelder–Mead has no general convergence guarantee: there are smooth convex functions on which it stops at a nonstationary point (McKinnon, 1998);
- noise can disturb the ordering of the vertices.

Derivative-free $\neq$ robust to noise.

KEY TAKEAWAYS — 3.2.3

1. Nelder–Mead uses a simplex of $n + 1$ points and only values of f .
2. At each iteration it identifies the worst vertex and searches on the opposite side: reflection.
3. The quality of the reflection decides: expand, accept, contract or shrink; the error is measured by Q .
4. The simplex adapts position, size and shape to the terrain, but the method is local. Nelder–Mead has no general convergence guarantee: there are smooth convex functions on which it stops at a nonstationary point (McKinnon, 1998).
5. Using only values of f does not mean being robust to noise.

Next (3.2.4): the simplex changes shape according to the values of f . What if, instead, we use a fixed pattern of points around the current point — the vertices of a box — and only move or shrink it? — Box's method.

References Full list: *Course bibliography*, at the end of the course PDF.

- Nelder, J. A., Mead, R., *A simplex method for function minimization*, The Computer Journal 7(4):308–313, 1965.
- McKinnon, K. I. M., *Convergence of the Nelder–Mead simplex method to a nonstationary point*, SIAM Journal on Optimization 9(1):148–158, 1998.
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009 — Ch. 8.

Resources

- MATLAB: `fminsearch`, `optimset('TolX',..., 'TolFun',..., 'Display', 'iter')`.
- Python: `scipy.optimize.minimize(f, x0, method='Nelder-Mead')(r.nfev)`.